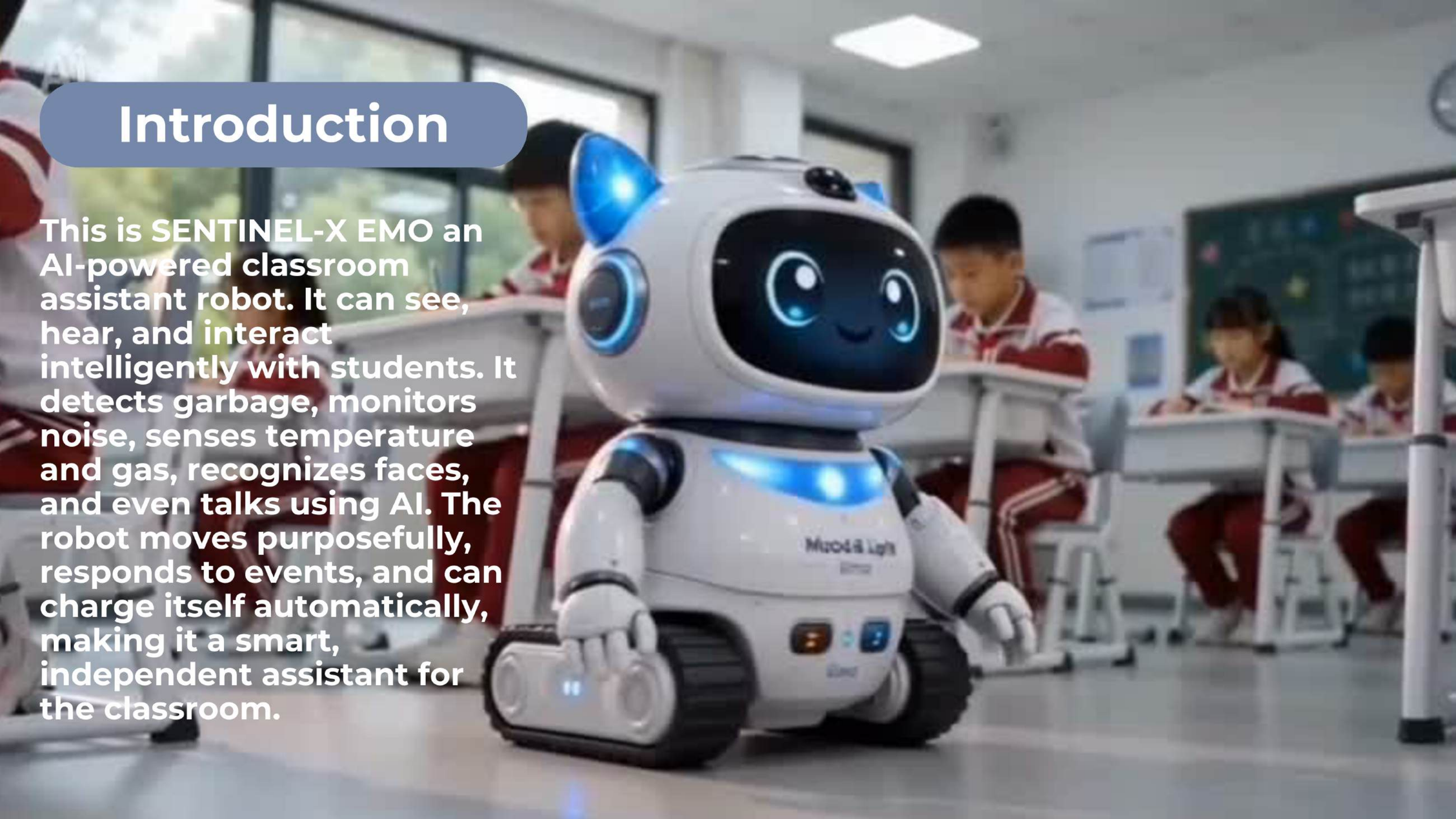
SENTINEL-X EMO

Watching, Learning, Protecting — Your Classroom’s Smartest Guardian

Introduction

This is SENTINEL-X EMO an AI-powered classroom assistant robot. It can see, hear, and interact intelligently with students. It detects garbage, monitors noise, senses temperature and gas, recognizes faces, and even talks using AI. The robot moves purposefully, responds to events, and can charge itself automatically, making it a smart, independent assistant for the classroom.



AIM

To create an intelligent, conversational, self-sustaining classroom robot that can understand speech, recognize students, reason using a large language model, and recharge itself automatically.

Objectives

- Connect ChatGPT / Gemini API via ESP32 WiFi for real-time, context-aware responses.
- Use INMP441 microphone + Whisper STT API for speech-to-text input.
- Replace pre-recorded audio with MAX98357A TTS DAC for natural speech output.
- Run face_recognition + YOLOv5-tiny on Raspberry Pi Zero W to identify students and detect actual garbage.
- Implement autonomous self-charging with IR homing, spring pins, TP4056 charger, and INA219 battery monitor.
- Demonstrate the complete v2.0 system live within a budget of Rs. 50,000

PROBLEM:

Traditional classroom robots aren't intelligent—they can't understand speech, recognize students, adapt, or monitor classroom conditions. They move randomly and need manual charging.

Proposed Solution

SENTINEL-X EMO

An AI-powered robot that recognizes students, remembers behaviour, interacts with voice, detects garbage, noise, gas, temperature, and humidity, moves purposefully, self-charges, and can optionally take attendance.

BLOCK DIAGRAM

INPUT LAYER

- **Ultrasonic** — Distance
- **MQ-2** — Gas-Smoke
- **DHT11** — Temp/Humidity
- **Touch** — Human detect
- **MPU-6050** — Tilt / Motion
- **INA219** — Battery monitor
- **IR Sensor** — Dock align
- **INMP441** — Microphone

PROCESSING LAYER

Raspberry Pi
Zero W — Vision AI
Face recog + YOLOv5

ESP32
Main Controller
WiFi · Sensors · Motors

ChatGPT / Gemini
AI API via WiFi
Language + reasoning

OUTPUT LAYER

OLED Face
Emotions

Displays real-time facial expressions such as happy, sad, angry and surprised

Reacts to sensor data and conversation context automatically

TTS DAC
Real voice

Converts AI text responses into clear natural spoken audio output

Uses different voice tones for alerts, greetings and AI conversations

Servo Arm
Gesture

Performs physical gestures such as waving, pointing and greeting movements

Triggered by face recognition results and touch sensor commands

DC Tracks
Movement

Drives the robot forward, backward and turns in all directions

Navigates autonomously using ultrasonic sensor obstacle data

RGB LED
Mood light

Changes color to reflect the robot's current mood and emotional state

Provides visual alerts for gas detection, low battery and danger events

Charge Dock
Auto-park

Automatically aligns with charging station using the IR sensor

Activates when INA219 battery monitor detects low charge level

Attendance Sheet
Auto-marking

Automatically marks student attendance by matching face recognition data to a student database

Logs time, date and student name into a digital attendance record stored and exportable as a sheet

Conversation Log
AI record

Saves all voice conversations between students and the robot via ChatGPT or Gemini

Helps teachers review questions asked and topics discussed by students with the robot

Student Report Card
Summary

Calculates each student's attendance percentage from the collected attendance data

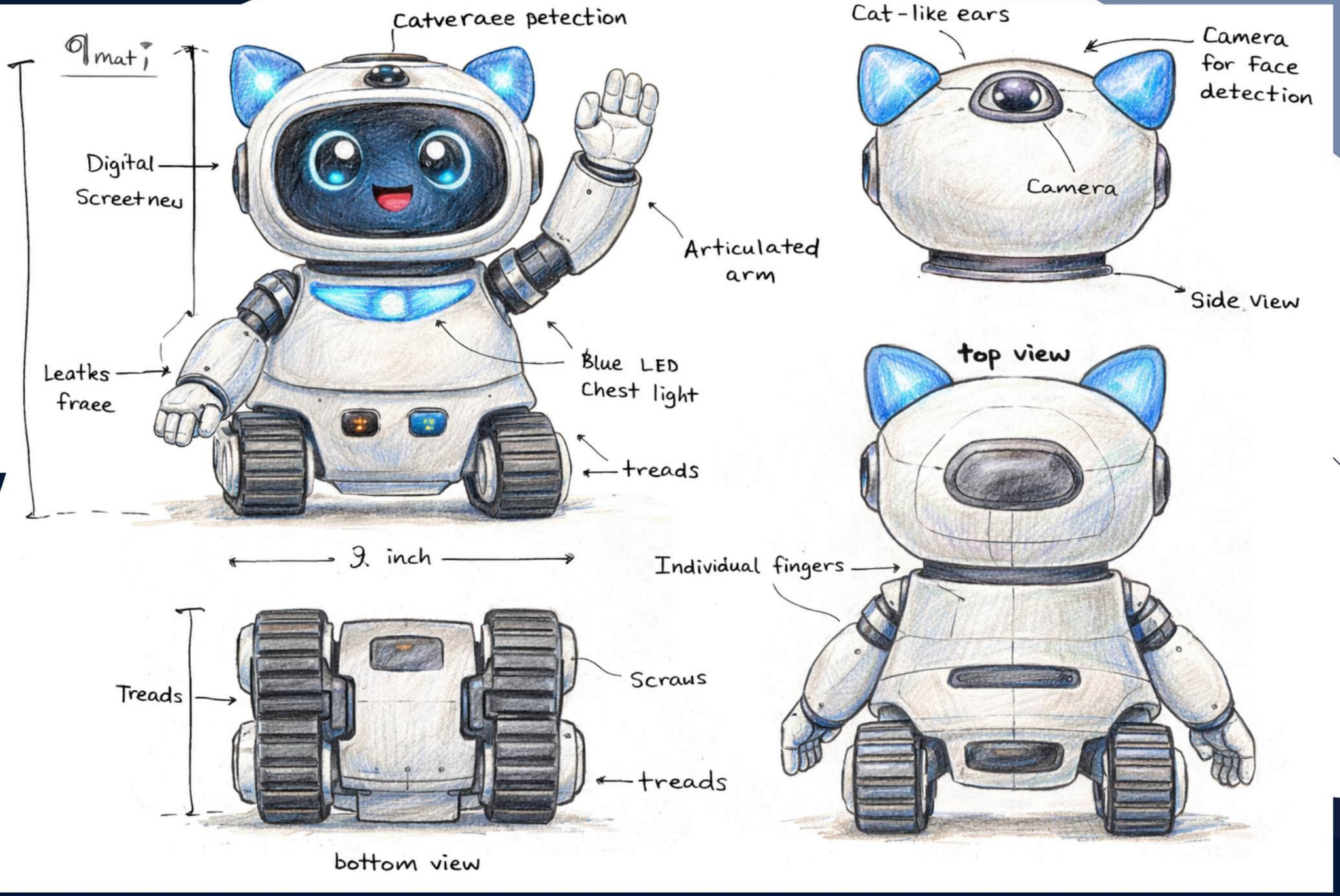
Generates a per-student summary card showing presence, absence and late records

Daily Activity Log
Full record

Records all robot events in one timeline — who it saw, alerts triggered, movements made

Gives a complete picture of the robot's daily operation useful for testing and review

3D VIEW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Digital Screetnew
Leatks fraee
Articulated arm
Blue LED Chest light
treads
9. inch
Cat-like ears
Camera for Face detection
Camera
Side View
top view
Individual fingers
Treads
Scraus
treads
bottom view

COMPONENTS

- Ultrasonic HC-SR04 – Distance Sensor
- MQ-2 – Gas/Smoke Sensor
- Pi Camera Module v2 + YOLOv5 – Vision / Object Detection
- DHT11 – Temperature Sensor
- DHT11 – Humidity Sensor
- Capacitive Touch – Touch Sensor
- MPU-6050 – IMU / Tilt Sensor
- INMP441 – Microphone (I2S Digital)
- MAX98357A – Audio DAC / Speaker Driver (I2S)
- Raspberry Pi Zero W – Microcomputer (AI Processing)
- INA219 – Battery Voltage / Current Sensor (I2C)
- TSOP4838 IR Receivers ×2 – IR Receiver
- IR LED Emitters ×2 – IR Transmitter
- TP4056 Charger IC – Li-Ion Charger
- Spring Contact Pins ×2 – Dock Connector
- USB-C 5V 3A Adapter – Power Supply
- ESP32 Dev Board – Microcontroller
- OLED 0.96" Display – Display (I2C)
- Servo Motor SG90 – Servo Motor
- DC Geared Motors ×2 – DC Motor
- L298N Motor Driver – Motor Driver (H-Bridge)
- RGB LED Ring (WS2812B) – LED Lighting
- Li-Ion Battery 7.4V 2000mAh – Battery
- Robot Chassis + Rubber Tracks – Mobile Platform

TECHNOLOGIES

- Arduino IDE / C++ (ESP32)
- Python 3 (Raspberry Pi Zero W)
- OpenAI Whisper API
- ChatGPT API (GPT-3.5-turbo)
- Online TTS (Google TTS / ElevenLabs)
- PWM (LEDC)
- I2C bus
- I2S bus
- UART

EXPECTED BUDGET

Component	Qty	Unit Price (LKR)	Total (LKR)
ESP32 Dev Board	1	3,200.00	3,200.00
Raspberry Pi Zero W	1	3,499.00	3,499.00
Pi Camera Module v2	1	7,950.00	7,950.00
INMP441 I2S Microphone	1	950.00	950.00
MAX98357A TTS DAC	1	850.00	930.00
INA219 Battery Sensor	1	570.00	570.00
TSOP4838 IR Receivers ×2	2	220.00	440.00
IR LED Emitters ×2 (Dock)	2	210.00	420.00
TP4056 Charger Module	1	250.00	250.00
Spring Contact Pins ×2	2	878.00	1,756.00
USB-C 5V 3A Adapter	1	664.00	664.00
Dock Frame + Copper Pads	1	1,600.00	1,600.00
OLED Display 1.54" (I2C)	1	2850.00	2850.00
Ultrasonic HC-SR04	1	625.00	625.00
MQ-2 Gas/Smoke Sensor	1	420.00	420.00
DHT11 Temp/Humidity Sensor	1	435.00	435.00
Capacitive Touch Module	1	149.00	149.00
MPU-6050 IMU Sensor	1	680.00	680.00
Servo Motor SG90	1	1,942.00	1,942.00
DC Geared Motors ×2 (6V)	2	2,084.00	4,168.00
L298N Motor Driver	1	890.00	890.00
Robot Chassis + Rubber Tracks	1	6,131.00	6,131.00
RGB LED Ring WS2812B (12 LEDs)	1	789.00	789.00
Li-Ion Battery 7.4V 2000mAh	1	1,433.00	1,433.00
Speaker 3W 8Ω	1	3,764.00	3,764.00
Wires, PCB, Misc. (incl. MicroSD)	-	-	3,620.00
TOTAL			**50,125.00**

WORK DISTRIBUTION

Omal (255514T AI) — Raspberry Pi Vision System

Hardware Responsibilities:

- Raspberry Pi Zero W setup & OS configuration
- Pi Camera Module v2 mounting
- UART communication wiring (Pi ↔ ESP32)
- MicroSD card setup
- Power rail integration

Software Responsibilities:

- Python face_recognition system
  - Student enrollment
  - 128-D embedding JSON database
- Real-time face recognition (every 2 seconds)
- YOLOv5-tiny garbage detection (~5 FPS, ~85% accuracy)
- UART JSON communication to ESP32
  - Example:
    - { "name": "Kasun", "confidence": 0.82 }
    - { "garbage": true, "class": "paper" }

Archana (254218G IT) — Emotion Display, Sensors & Integration

Hardware Responsibilities:

- OLED 0.96" I2C display setup
- WS2812B RGB LED ring (12 LEDs)
- SG90 servo motor (arm gestures)
- MQ-2 smoke sensor
- PIR motion sensor
- DHT11 temperature & humidity sensor
- LDR light sensor
- MPU-6050 IMU
- Sound sensor module

Software Responsibilities:

- OLED emotion animations
- (Happy, Sad, Angry, Scared, Sleepy, Surprised, Charging, etc.)
- RGB LED emotion colour mapping
- Servo gesture programming
- Sensor threshold logic (all sensors)
- Full system integration & testing coordination

WORK DISTRIBUTION

Suvin Sathsara (255035B ITM) — Self-Charging Auto-Dock System

Hardware Responsibilities:

- INA219 battery monitor (I2C wiring)
- TSOP4838 IR receivers (front-left & front-right mounting)
- IR LED emitters on dock frame
- Spring-loaded gold contact pins
- TP4056 charging circuit
- Copper contact pads on docking station
- USB-C 5V/3A adapter wiring
- Dock frame physical construction

Software Responsibilities:

- Battery state machine
- (Patrol → Low → Seek → Dock → Charge → Resume)
- IR homing alignment logic
- INA219 I2C polling (every 30 seconds)

Sanduni (254140J IT) — AI Voice Pipeline (STT → ChatGPT → TTS)

Hardware Responsibilities:

- INMP441 I2S microphone mounting & wiring
- MAX98357A DAC wiring
- 3W 8Ω speaker installation
- Capacitive touch sensor (wake-word trigger)

Software Responsibilities:

- I2S audio capture on ESP32
- Whisper STT API integration (HTTP POST)
- ChatGPT API integration with context (student + sensors)
- Google TTS / ElevenLabs audio streaming
- Audio playback via MAX98357A
- ChatGPT response sentiment parsing → emotion trigger

WORK DISTRIBUTION

Rishma (255009B ITM) — Chassis, Drive System & Power

Hardware Responsibilities:

- Robot chassis assembly
- Rubber tank tracks installation
- DC geared motors ×2 setup
- L298N motor driver wiring
- 7.4V Li-Ion battery integration
- Main power distribution board setup
- Complete wiring harness management

Software Responsibilities:

- ESP32 PWM motor control using LEDC
- Obstacle avoidance logic (HC-SR04)
- Differential drive (skid-steer) algorithm

REFERENCES

[1] Espressif Systems. (2023). ESP32 Technical Reference Manual v5.1.
https://www.espressif.com/en/support/documents/technical-documents

[2] Raspberry Pi Foundation. (2024). Raspberry Pi Zero 2 W Documentation.
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[3] OpenAI. (2024). Whisper API and ChatGPT API Documentation.
https://platform.openai.com/docs

[4] Geitgey, A. (2020). face_recognition Python Library.
https://github.com/ageitgey/face_recognition

THANK YOU